

Proposal: Relationship between Real Estate Loans and Mortgage rate (US) and Forecasting

Main Question:

How does mortgage rate change that affect the volume of real estate loans from all commercial banks in US. Specifically, we aim to understand how these two variables are related over the period from 1980 to 2023. There are two variable we should include in this model, CPI and investment rate.

Main Reference:

Grégory Levieuge (2017) Explaining and forecasting bank loans. Good times and crisis, *Applied Economics*, 49:8, 823-843, DOI:10.1080/00036846.2016.1208350

Main Econometric Technique:

VECM (vector error-correction models), VAR (vector auto-regressive)

Data:

All data source of this project is from Fred website. In this model, we have Real Estate Loans of All Commercial Banks (Monthly, Percent Change), 30-Year Fixed Rate Mortgage Average in the United States (Monthly, Average), CPI (All Items) and Investment rate. From the reference paper, author use 3-month bonds rate as investment rate in Euro area, we don't have 3-month bonds rate in US, I have Short-Term 3-month or 90-day Deposit Rates Yields (Percent) instead.